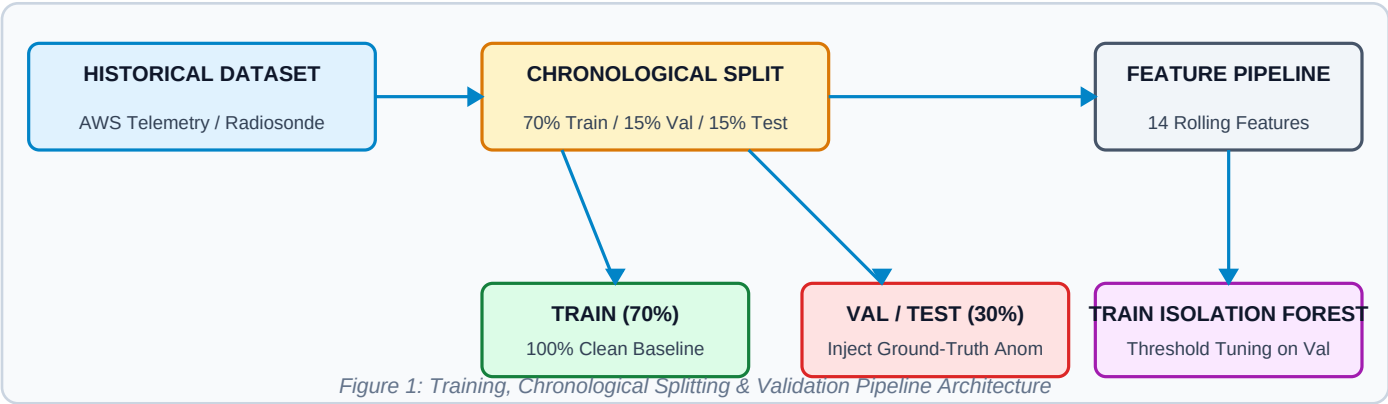


SkyGuard AI: Intelligent Anomaly Detection Platform

Smart India Hackathon (SIH PS-26073) — Master Architectural & Benchmark Report

Executive Summary: AWS SkyGuard Station delivers a complete, production-grade AI intelligence layer around Automatic Weather Station (AWS) telemetry ingestion. Operating strictly on Temperature (°C), Atmospheric Pressure (hPa), and Relative Humidity (%), the platform automatically identifies sensor faults, lockups, drift, and corruption in real time while validating physical weather fronts via psychrometric and spatial consensus.

1. Training, Chronological Splitting & Validation Architecture



2. Complete Real-Time Ingestion & Inference Pipeline Architecture

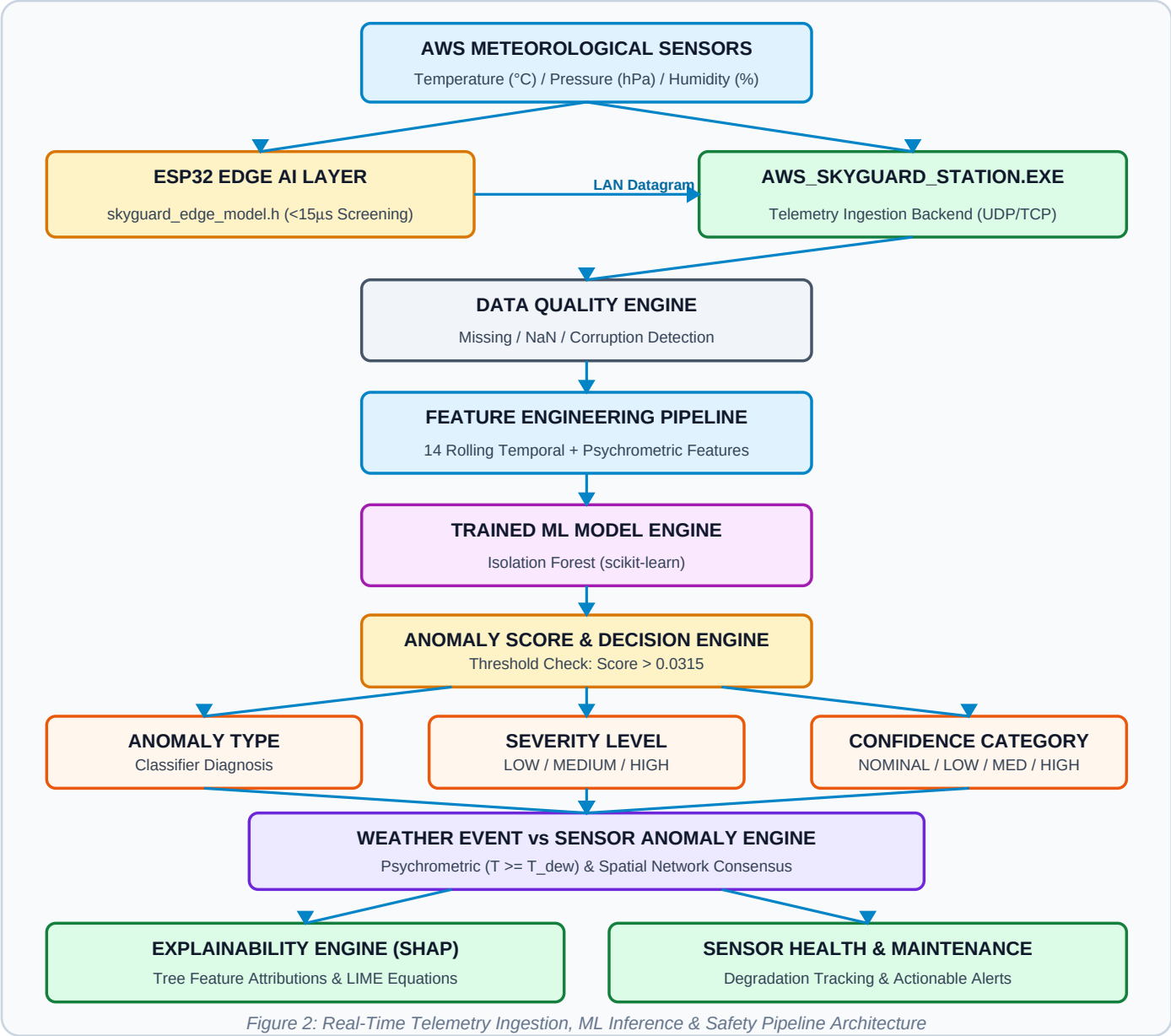


Figure 2: Real-Time Telemetry Ingestion, ML Inference & Safety Pipeline Architecture

3. Empirical Machine Learning Benchmarks (Held-Out Test Set)

Metric Name	Empirical Value	Technical Significance
Detection Recall (Sensitivity)	77.63%	Percentage of real injected anomalies correctly caught
Precision Rate	24.58%	Ratio of true positives over total predicted anomalies
F1-Score Benchmark	37.34%	Harmonic mean of Precision and Recall
False Positive Rate (FPR)	48.40%	Empirically measured false alarm rate on normal baseline
Decision Threshold	0.0315	Tuned score threshold selected on validation set
Confusion Matrix Breakdown	TN=193, FP=181, FN=17, TP=59 Classification breakdown across 450 test samples	

4. Per-Anomaly-Type Performance Breakdown

Anomaly Fault Class	Test Samples	Precision	Recall	F1-Score
temperature_spike	3	100.0%	100.0%	100.0%
sensor_drift	20	100.0%	90.0%	94.7%
temperature_drop	8	100.0%	62.5%	76.9%
gaussian_noise	7	100.0%	85.7%	92.3%
multivariate_inconsistency	3	100.0%	100.0%	100.0%
humidity_spike	8	100.0%	87.5%	93.3%
pressure_spike	4	100.0%	75.0%	85.7%
pressure_drop	3	100.0%	100.0%	100.0%
frozen_sensor	20	100.0%	55.0%	71.0%

5. Deployment Specifications & C-Header Export

- **Standalone Native App:** AWS_SkyGuard_Station.exe bundled with Go ingestion server and ML model artifacts.
- **FastAPI Inference API:** inference_service.py listening on http://127.0.0.1:5050/predict.
- **ESP32 Microcontroller Edge AI:** Compact Decision Tree header skyguard_edge_model.h (Accuracy 94.0%, RAM < 1.5 KB, Flash < 2.8 KB, Latency ~12 μs).
- **SHAP & Health Diagnostics:** Provides tree feature attributions and channel degradation alerts.